

ISC Class XII Mathematics

Complete Formula Reference

Chapters 1 – 8

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CHAPTER 1

Relations & Functions

Formula Reference Sheet

ISC Class XII Maths — Chapter 1

Relations & Functions — Formula Sheet

A. Relations — Definitions & Counting

$$n(R) = 2^{n^2} \quad n(R_{\text{reflexive}}) = 2^{n^2 - n}$$

Reflexive: $(a, a) \in R \quad \forall a \in A$

Symmetric: $(a, b) \in R \implies (b, a) \in R$

Transitive: $(a, b), (b, c) \in R \implies (a, c) \in R$

Equivalence relation = reflexive + symmetric + transitive

Equivalence class: $[a] = \{x \in A : (x, a) \in R\}$

Tip: Test *reflexive* first (every diagonal pair (a, a) must appear). For *symmetric* and *transitive*, don't try to prove they hold everywhere — actively hunt for ONE counter-example pair; a single failure is enough to disprove either property.

B. Domain & Range Rules

Even root: $g(x) \geq 0$ **Denominator** $\neq 0$

$\log(g(x))$ defined $\iff g(x) > 0$

$$a \sin x + b \cos x = r \sin(x + \alpha), \quad r = \sqrt{a^2 + b^2}$$

Tip: Auxiliary angle method: for $a \sin x + b \cos x$, put $a = r \cos \alpha$, $b = r \sin \alpha$ where $r = \sqrt{a^2 + b^2}$. Since $-1 \leq \sin(x + \alpha) \leq 1$, the range is exactly $[-r, r]$. (A single term like $5 \sin 3x + 2$ needs no auxiliary angle — just shift the standard $[-1, 1]$ range.)

C. Standard Functions — Domain/Range

Identity: $f(x) = x$ Dom = $\mathbb{R}$, Range = $\mathbb{R}$

Constant: $f(x) = c$ Dom = $\mathbb{R}$, Range = $\{c\}$

Modulus: $f(x) = |x|$ Dom = $\mathbb{R}$, Range = $[0, \infty)$

Signum: $f(x) = \frac{|x|}{x}$ ($x \neq 0$), Range = $\{-1, 0, 1\}$
 $f(0) = 0$

Greatest integer: $f(x) = [x]$ Dom = $\mathbb{R}$, Range = $\mathbb{Z}$

D. Types of Functions — Tests & Counting

$$f \text{ one-one} \iff f(x_1) = f(x_2) \implies x_1 = x_2$$

$$f: A \rightarrow B \text{ onto} \iff \forall y \in B, \exists x \in A : f(x) = y$$

Functions $A \rightarrow B = n^m$ ($m = |A|$, $n = |B|$)

One-one functions = ${}^n P_m$ ($m \leq n$); 0 ($m > n$)

Onto ($n=2$): $2^m - 2$; ($n=3$): $3^m - 3(2^m - 1)$

Bijections = $m!$ ($m = n$); 0 ($m \neq n$)

Tip: Fix which set is the domain (size m) and which is the codomain (size n) before substituting — it's easy to swap them by mistake. When $m = n$, a one-one function is automatically onto too, so it simplifies straight to $m!$.

E. Composition of Functions

$$(g \circ f)(x) = g(f(x))$$

$$f, g \text{ one-one} \implies g \circ f \text{ one-one}$$

$$f, g \text{ onto} \implies g \circ f \text{ onto}$$

Tip: $(g \circ f)(x) = g(f(x))$: work out $f(x)$ first, then plug that ENTIRE result into g . Composition is generally NOT commutative ($g \circ f \neq f \circ g$) — never assume, always compute both sides. If both composites simplify to plain x (i.e. $g \circ f = f \circ g = I$), the two functions are mutual inverses.

F. Invertible Functions

$$f \text{ invertible} \iff f \text{ bijective (one-one \& onto)}$$

$$\text{Put } y = f(x), \text{ solve for } x \implies f^{-1}(y) = x$$

$$(f^{-1})^{-1} = f$$

$$f \text{ self-inverse} \iff f^{-1}(x) = f(x) \quad (\text{check } f(f(x)) = x)$$

$$f^{-1} = \{(b, a) : (a, b) \in f\}$$

Restricting Domain to Invert

Quadratic on $\mathbb{N}$ (e.g. $f(x) = 4x^2 + 12x + 15$): complete the square first: $4x^2 + 12x + 15 = (2x + 3)^2 + 6$. Since $x \in \mathbb{N}$ makes $(2x+3) > 0$, only the *positive* root applies when inverting — this makes f one-one, and restricting the codomain to $S = \text{range}(f)$ makes it onto automatically. Then $f^{-1}(y) = \frac{\sqrt{y-6} - 3}{2}$.

Non-monotonic cubic (e.g. $f(x) = 2x^3 - 3x^2 - 12x + 8$): a wavy graph is never one-one on all of $\mathbb{R}$. Find turning points by solving $f'(x) = 0$ — these bound the LARGEST interval containing the required point on which f is monotonic (hence invertible). Then domain of $f^{-1} = \text{range of } f \text{ on that interval}$, and range of $f^{-1} = \text{the restricted interval itself}$.

G. Quick Recognition Guide

Pattern seen	Use
Check R/S/T of a relation	Test all 3 directly; hunt a counter-example for S, T
Prove equivalence relation + find class	Prove R,S,T for general elements, then test the condition against every element
Domain/range: surd, rational, modulus	Domain: root ≥ 0 , denom $\neq 0$; range: simplify fully, track excluded values
Domain: log/trig expression	Combine log > 0 , trig inequality, denom $\neq 0$ together
$a \sin x + b \cos x$ type	Auxiliary angle, range $[-r, r]$, $r = \sqrt{a^2 + b^2}$
Prove one-one	Assume $f(x_1) = f(x_2)$, solve to $x_1 = x_2$
Prove onto/not onto	Solve $f(x) = y$ for x ; check x lies in the domain
Counting functions/bijections	Pick right formula: n^m , ${}^n P_m$, onto shortcut, or $m!$
Composition of algebraic functions	Substitute the ENTIRE inner function into the outer
Prove mutual inverses	Show $g \circ f = f \circ g = I$
Find inverse (algebraic)	Set $y = f(x)$, solve for x
Invertibility from graph/diagram	2 arrows/points sharing a target $\Rightarrow$ not one-one; vertical line hits twice $\Rightarrow$ not even a function
Restrict codomain (quadratic on $\mathbb{N}$)	Complete the square, take positive root only
Restrict domain (non-monotonic function)	Find turning points via $f'(x) = 0$, take largest interval

CHAPTER 2

Inverse Trigonometric Functions

Formula Reference Sheet

ISC Class XII Maths — Chapter 2

Inverse Trigonometric Functions — Formula Sheet

A. Domains & Principal Value Branches

$$\sin^{-1} x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \cos^{-1} x : [0, \pi] \quad \tan^{-1} x : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cot^{-1} x : (0, \pi) \quad \sec^{-1} x : [0, \pi] \setminus \left\{\frac{\pi}{2}\right\} \quad \operatorname{cosec}^{-1} x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$$

Input domains: $\sin^{-1}, \cos^{-1}$ need $|x| \leq 1$; $\tan^{-1}, \cot^{-1}$ take all $x \in \mathbb{R}$; $\sec^{-1}, \operatorname{cosec}^{-1}$ need $|x| \geq 1$.

B. Inverse-Composition Identities

$f(f^{-1}(x)) = x$ whenever f^{-1} is defined at x (e.g. $\sin(\sin^{-1} x) = x, |x| \leq 1$).

$f^{-1}(f(\theta)) = \theta$ **only if** θ already lies in f^{-1} 's principal branch (Section A) — otherwise reduce θ into range first.

Tip: For $f^{-1}(f(\theta))$ with θ outside range: rewrite using a supplementary/periodic identity (e.g. $\sin(\pi - \theta) = \sin \theta$, $\cos(2\pi - \theta) = \cos \theta$, $\tan(\theta - \pi) = \tan \theta$) until the reduced angle sits inside the branch.

C. Negative-Argument Identities

$$\text{Odd: } \sin^{-1}(-x) = -\sin^{-1} x \quad \tan^{-1}(-x) = -\tan^{-1} x \quad \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

$$\text{Supplementary: } \cos^{-1}(-x) = \pi - \cos^{-1} x \quad \cot^{-1}(-x) = \pi - \cot^{-1} x \quad \sec^{-1}(-x) = \pi - \sec^{-1} x$$

D. Co-function Pairs

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}, |x| \leq 1 \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}, x \in \mathbb{R}$$

$$\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}, |x| \geq 1$$

E. Reciprocal / Conversion Identities

$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x, |x| \geq 1 \quad \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x, |x| \geq 1$$

$$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x, x > 0; \text{ for } x < 0: \tan^{-1}\left(\frac{1}{x}\right) = -\pi + \cot^{-1} x$$

F. Sum/Difference — Tangent Inverse

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1$$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), xy > -1$$

Tip: If $x, y > 0$ and $xy > 1$: add π to the RHS. If $x, y < 0$ and $xy > 1$: subtract π . **Triple sum:** $\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$ — use when three inverse-tangent terms add to a known angle.

G. Sum Formulas — Sine & Cosine

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right)$$

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right)$$

Tip: If the result falls outside the natural principal range, adjust by $\pi - (\cdot)$ or add π as the sign/quadrant of x, y demands.

H. Double-Angle (2x) Forms

$$2 \tan^{-1} x = \sin^{-1} \left(\frac{2x}{1+x^2} \right), |x| \leq 1$$

$$2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), x \geq 0$$

$$2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right), -1 < x < 1$$

Tip: These come from putting $x = \tan \theta$ into $\sin 2\theta, \cos 2\theta, \tan 2\theta$ — spot them whenever an expression contains $\frac{2x}{1+x^2}, \frac{1-x^2}{1+x^2}$, or $\frac{2x}{1-x^2}$.

Right-Triangle Conversion

To convert $\theta = \sin^{-1} \left(\frac{p}{h} \right)$ into another ratio:

1. Draw a right triangle: opposite = p , hypotenuse = h .
2. Adjacent = $\sqrt{h^2 - p^2}$ (Pythagoras).
3. Read off the needed ratio (e.g. $\tan \theta = \frac{p}{\sqrt{h^2 - p^2}}$).

Works the same starting from $\cos^{-1}$ or $\tan^{-1}$ — always label opposite/adjacent/hypotenuse first, then Pythagoras, then read the ratio.

Weierstrass Substitution

For expressions mixing $\sin \theta$ and $\cos \theta$ inside a $\tan^{-1}$:

1. Let $t = \tan \frac{\theta}{2}$.
2. Convert: $\sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$, $\tan \theta = \frac{2t}{1-t^2}$.
3. Substitute, simplify the resulting expression in t .
4. Back-substitute $t = \tan \frac{\theta}{2}$ in the final answer.

K. Half-Angle Trig (for simplification)

$$1 - \cos x = 2 \sin^2 \frac{x}{2} \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} \quad \tan \left(\frac{\pi}{4} - x \right) = \frac{1 - \tan x}{1 + \tan x}$$

Tip: Standard substitutions for $\tan^{-1}(f(\sin x, \cos x))$ -type simplification: $x = \sin \theta \Rightarrow \sqrt{1-x^2} = \cos \theta$; $x = \tan \theta \Rightarrow \sqrt{1+x^2} = \sec \theta$; $x = \cos \theta \Rightarrow \sqrt{1-x^2} = \sin \theta$; $x = a \sin \theta \Rightarrow \sqrt{a^2 - x^2} = a \cos \theta$.

M. Common Traps to Avoid

- $\cos^{-1}(-x) \neq -\cos^{-1} x$ — it equals $\pi - \cos^{-1} x$ (range is $[0, \pi]$, never negative).
- $f^{-1}(f(\theta)) = \theta$ only when θ is already inside the principal branch — always check first.
- In $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$, forgetting the $\pm\pi$ correction when $xy > 1$ silently flips the sign of the answer.
- When solving equations, always verify each root against the *original* domain — squaring or combining steps can introduce extraneous roots.
- $\sec^{-1} x$ and $\operatorname{cosec}^{-1} x$ are undefined for $-1 < x < 1$; don't apply the reciprocal identities there.

L. Quick Recognition Guide

Pattern seen	Use
$f^{-1}(f(\theta))$, θ outside branch	Sec. B — reduce θ first
$\cos(\sin^{-1} x)$, $\tan(\cos^{-1} x)$, etc.	Right-Triangle Method
$f(-x)$ inside inverse trig	Sec. C odd/supplementary
Two inverse terms summing	Sec. D (co-function) or F/G
$\frac{1}{x}$ inside inverse trig	Sec. E reciprocal identity
$xy > 1$ in $\tan^{-1} x + \tan^{-1} y$	Sec. F, add/subtract π
Contains $\frac{2x}{1+x^2}$, $\frac{1-x^2}{1+x^2}$	Sec. H double-angle form
$\sqrt{\frac{1-\cos x}{1+\cos x}}$ -type	Sec. K half-angle $\rightarrow$ Weierstrass
Solve for x : sum of two inverse terms = const	Combine (F/G), then invert
Three inverse-tangent terms sum to known angle	Sec. F triple-tangent identity
Given $x-y-xy$ or similar from a difference equation	Sec. F, isolate via $\tan$ of both sides
Find $\max/\min$ value of $(\text{inverse trig})^k$	Sec. A — use endpoint of principal range

CHAPTER 3

Matrices

Formula Reference Sheet

ISC Class XII Maths — Chapter 3

Matrices — Formula Sheet

A. Order, Types & Counting

Matrix of order $m \times n$ has mn elements, general element a_{ij} (i = row, j = column).

Row matrix: $1 \times n$ **Column matrix:** $m \times 1$
Square matrix: $m = n$

Diagonal: square, $a_{ij} = 0$ for $i \neq j$ **Scalar:** diagonal with all a_{ii} equal **Identity I :** diagonal, all $a_{ii} = 1$

Total number of $m \times n$ matrices possible with each entry chosen from k given values = k^{mn} .

B. Equality of Matrices

$A = [a_{ij}]$ and $B = [b_{ij}]$ are equal $\iff$ same order
and $a_{ij} = b_{ij}$ for every i, j .

Tip: Equate corresponding entries one pair at a time — this turns a matrix equation into a system of ordinary algebraic equations you already know how to solve.

C. Addition, Subtraction & Scalar Mult.

$A \pm B = [a_{ij} \pm b_{ij}]$ (only if A, B have the same order)
 $kA = [k a_{ij}]$

$kA + mB = [k a_{ij} + m b_{ij}]$

Tip: Addition is commutative and associative: $A + B = B + A$, $(A + B) + C = A + (B + C)$. Use this freely to regroup terms when isolating an unknown matrix.

D. Multiplication — Compatibility

$P_{m \times n} \cdot Q_{n \times p} = R_{m \times p}$ — defined **only if** inner dimensions match ($n = n$); result takes the outer dimensions.

$(AB)_{ij} = \sum_k a_{ik} b_{kj} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B)$

E. Properties of Multiplication

Associative: $(AB)C = A(BC)$ **Distributive:** $A(B + C) = AB + AC$

Identity: $AI = IA = A$ **Generally NOT commutative:** $AB \neq BA$

Tip: Never cancel a common matrix factor or assume $AB = BA$ unless it is proved for that specific pair. $AB = O$ does *not* imply $A = O$ or $B = O$.

Reducing Matrix Polynomials

Given a relation like $A^2 = 4A - 7I$ (from $A^2 - 4A + 7I = O$):

1. Write the higher power as a product: $A^3 = A \cdot A^2$.
2. Substitute the given relation for A^2 , expand.
3. Reduce again if an A^2 term reappears, until only A and I remain.
4. Substitute the actual matrix entries for A and I only at the *final* step.

Induction on Matrices

To prove $P(n) : A^n = (\text{formula in } n)$ for all $n \in \mathbb{N}$:

1. **Base case** — verify $P(1)$: check A^1 equals the formula at $n = 1$.
2. **Inductive hypothesis** — assume $P(k)$ true: $A^k = (\text{formula in } k)$.
3. **Inductive step** — write $A^{k+1} = A^k \cdot A$, substitute the hypothesis, multiply out, and show it equals the formula at $n = k + 1$.
4. Conclude by the Principle of Mathematical Induction.

F. Solving Matrix Equations

Linear combinations (e.g. $2X + Y = B$, $X + Y = A$, $X - Y = B$): treat X, Y as whole matrices and eliminate/substitute exactly as with simultaneous linear equations, *then* solve entry-by-entry if needed.

Multiplicative (e.g. find B given $BA = I$): let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, multiply out, then equate entries to get simultaneous equations in a, b, c, d .

G. Transpose Properties

$$(A')' = A \quad (kA)' = kA' \quad (A \pm B)' = A' \pm B'$$

$$\text{Reversal law: } (AB)' = B'A' \quad (ABC)' = C'B'A'$$

Tip: The reversal law flips *both* the order of multiplication and which factor is transposed — the single most common slip is writing $(AB)' = A'B'$ instead of $B'A'$.

H. Symmetric & Skew-Symmetric

Symmetric: $A' = A \iff a_{ij} = a_{ji}$ for all i, j (mirror about the leading diagonal).

Skew-symmetric: $A' = -A \iff a_{ij} = -a_{ji}$; in particular $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$ — every diagonal entry is 0.

Symmetric + Skew Decomposition

Every square matrix A splits uniquely as $A = P + Q$ where

$$P = \frac{1}{2}(A + A') \text{ is } \mathbf{symmetric} \quad (\text{check: } P' = \frac{1}{2}(A' + A) = P)$$

$$Q = \frac{1}{2}(A - A') \text{ is } \mathbf{skew-symmetric} \quad (\text{check: } Q' = \frac{1}{2}(A' - A) = -Q)$$

Verify $P + Q = A$ as the final check.

I. Orthogonal & Rotation Matrices

A is **orthogonal** if $A'A = AA' = I$.

Rotation form $M(\theta) = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ satisfies $M(x)M(y) = M(x + y)$.

Standard identity used throughout: $\sin^2 \theta + \cos^2 \theta = 1$, and $\cos A \cos B \mp \sin A \sin B = \cos(A \mp B)$, $\sin A \cos B \pm \cos A \sin B = \sin(A \pm B)$.

J. Common Traps to Avoid

- Check order compatibility *before* multiplying — AB can exist while BA does not.
- $(AB)' = B'A'$, never $A'B'$ — the reversal law reverses order.
- $A^2 = A \cdot A$ using *matrix* multiplication, never entry-wise squaring.
- In induction, the inductive step must start from $A^{k+1} = A^k \cdot A$ (or $A \cdot A^k$) — don't jump straight to the claimed formula.
- A skew-symmetric matrix always has zero diagonal — use this as a quick sanity check.

K. Quick Recognition Guide

Pattern seen	Use
Matrix built from $a_{ij} = f(i, j)$	Sec. A — compute entry by entry
Two matrices set equal	Sec. B — equate entries
Sum/difference/ kA of matrices	Sec. C
Product of two matrices, or find one entry of it	Sec. D entry formula
Prove $(AB)C = A(BC)$ or $A(B + C) = AB + AC$	Sec. E properties
Given $A^2 =$ linear in A, I ; find $A^3, A^4, \dots$	Worked Pattern: Polynomial Reduction
“Prove $A^n = \dots$ for all n ”	Worked Pattern: Induction
Find unknown matrix X from linear equation(s)	Sec. F linear combination
Find B given $BA = I$ or similar	Sec. F multiplicative, assume entries
Any question with $(\cdot)'$ transpose in it	Sec. G
“Show matrix is symmetric/skew-symmetric”	Sec. H definition
“Express A as sum of symmetric + skew”	Worked Pattern: Decomposition
“Verify $A'A = I$ ” or rotation matrix $M(\theta)$	Sec. I orthogonal/rotation

CHAPTER 4

Determinants

Formula Reference Sheet

ISC Class XII Maths — Chapter 4

Determinants — Formula Sheet

A. Determinant Basics

$$2 \times 2: \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$3 \times 3 \text{ (expand along } R_1): \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Minor M_{ij} : determinant left after deleting row i , column j . **Cofactor** $A_{ij} = (-1)^{i+j} M_{ij}$

Expansion by cofactors (any row/column): $\Delta = a_{i1}A_{i1} + a_{i2}A_{i2} + a_{i3}A_{i3}$

B. Row/Column Operations

Swap $R_i \leftrightarrow R_j$ (or $C_i \leftrightarrow C_j$): determinant **changes sign**.

$R_i \rightarrow R_i + kR_j$ (or column analog): determinant **unchanged**.

Two identical/proportional rows or columns $\Rightarrow \Delta = 0$.

Common factor k in one row/column can be taken

$$\text{outside: } \begin{vmatrix} ka_1 & b_1 & c_1 \\ ka_2 & b_2 & c_2 \\ ka_3 & b_3 & c_3 \end{vmatrix} = k \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

C. Algebraic Properties

$$|A^T| = |A| \quad |AB| = |A||B| \quad |A^n| = |A|^n$$

$|kA| = k^n|A|$ for $n \times n$ matrix A (*not* $k|A|$ — every row scales) $|I| = 1$

Tip: $|kA| = k^n|A|$ scales by k in *every* row, so the power is n (matrix size), e.g. $|2A| = 2^3|A| = 8|A|$ for a 3×3 matrix. Confusing this with $k|A|$ is the most common slip in this chapter.

D. Standard Determinant Identities

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$\begin{vmatrix} 1 & a & a^3 \\ 1 & b & b^3 \\ 1 & c & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$$

Same layout with a^3 swapped in for the a^2 column: $(a-b)(b-c)(c-a)(ab+bc+ca)$

$$\begin{vmatrix} x & y & z \\ x^2 & y^2 & z^2 \\ x^3 & y^3 & z^3 \end{vmatrix} = xyz \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Row/Column Sum Trick

When every row (or column) shares a common sum (e.g. each row adds to $a+b+c$):

1. Apply $C_1 \rightarrow C_1 + C_2 + C_3$ (or $R_1 \rightarrow R_1 + R_2 + R_3$) to pull that common sum into one column/row.
2. Take the common factor outside the determinant.
3. Simplify remaining rows/columns with further $R_i \rightarrow R_i - R_j$ operations — zeros usually appear, making expansion easy.

E. Area of Triangle & Collinearity

Area of triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Take the **absolute value** (area can't be negative).

Collinear $\iff$ this determinant = 0 (area = 0).

Tip: Equation of line through $(x_1, y_1), (x_2, y_2)$: any point (x, y) on the line satisfies $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$ — expand along R_1 to get the line's equation directly.

F. Singular Matrices & Adjoint

A is **singular** $\iff |A| = 0$ (non-singular $\iff |A| \neq 0$).

$\text{adj } A = C'$ where $C = [C_{ij}]$ is the cofactor matrix (transpose of cofactors).

2×2 **shortcut**: for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\text{adj } A = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$
— swap diagonal, negate off-diagonal.

G. Key Adjoint/Inverse Identities

$A(\text{adj } A) = (\text{adj } A)A = |A|I \quad A^{-1} = \frac{1}{|A|} \text{adj } A$
(needs $|A| \neq 0$)

$|\text{adj } A| = |A|^{n-1}$ for $n \times n$ (so $= |A|^2$ when $n = 3$)
 $|A^{-1}| = \frac{1}{|A|}$

$\text{adj}(\text{adj } A) = |A|^{n-2}A$ for $n \times n$

H. Inverse Properties & Reversal Laws

$AA^{-1} = A^{-1}A = I \quad (A^{-1})^{-1} = A \quad (A^T)^{-1} = (A^{-1})^T$

Reversal laws: $(AB)^{-1} = B^{-1}A^{-1} \quad \text{adj}(AB) = (\text{adj } B)(\text{adj } A)$

If $AB = kI$ then $A^{-1} = \frac{1}{k}B$.

Inverse from Matrix Polynomial

Given $A^2 - c_1A - c_0I = O$ with $c_0 \neq 0$:

1. Rearrange: $A^2 - c_1A = c_0I$.
2. Factor out A : $A(A - c_1I) = c_0I$.
3. Divide by c_0 : $A^{-1} = \frac{1}{c_0}(A - c_1I)$.

Same idea for a cubic relation $A^3 - c_2A^2 - c_1A - c_0I = O$:
 $A^{-1} = \frac{1}{c_0}(A^2 - c_2A - c_1I)$.

Solving $AX = B$

Also called *Martin's Rule*.

1. Write the system as $AX = B$ (coefficient matrix $\times$ variable column = constant column).
2. Compute $|A|$; if $|A| \neq 0$, find $\text{adj } A$.
3. $X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$.
4. Read off each variable from the resulting column matrix X .

Tip: If the equations contain $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ instead of x, y, z :
substitute $u = \frac{1}{x}, v = \frac{1}{y}, w = \frac{1}{z}$ first to make the system linear in u, v, w , solve $AU = B$ as above, then **back-substitute** $x = \frac{1}{u}, y = \frac{1}{v}, z = \frac{1}{w}$ — don't leave the answer in terms of u, v, w .

I. Consistency of $AX = B$

$|A| \neq 0 \Rightarrow$ **unique solution** (consistent), $X = A^{-1}B$.

$|A| = 0$ and $(\text{adj } A)B \neq O \Rightarrow$ **no solution** (inconsistent).

$|A| = 0$ and $(\text{adj } A)B = O \Rightarrow$ **infinitely many solutions**.

J. Homogeneous Systems $AX = O$

$|A| \neq 0 \Rightarrow$ **only the trivial solution** $x = y = z = 0$.

$|A| = 0 \Rightarrow$ **infinitely many non-trivial solutions** (always consistent — $X = O$ is always one solution).

K. Common Traps to Avoid

- $|kA| = k^n|A|$, **not** $k|A|$ — every row is scaled by k .
- Swapping two rows/columns flips the *sign* of Δ — easy to forget mid-proof.
- A^{-1} exists only when $|A| \neq 0$ — always state this before writing $A^{-1} = \frac{1}{|A|} \text{adj } A$.
- For $|A| = 0$: don't assume "infinitely many solutions" automatically — check $(\text{adj } A)B$ first; it could be inconsistent instead.
- A homogeneous system is *never* inconsistent — $|A| = 0$ always gives infinite (non-trivial) solutions, never "no solution."

L. Quick Recognition Guide

Pattern seen	Use
Plain $2 \times 2/3 \times 3$ evaluate	Sec. A expansion
Determinant with a, b, c symmetric, rows share a sum	Worked Pattern: Row/Col Sum Trick
Prove $\Delta =$ product like $(a-b)(b-c)(c-a)$	Sec. D standard identity
“Find area” / “show collinear”	Sec. E
“Find equation of line through two points”	Sec. E line-through-two-points determinant
$ A = 0$ given/asked	Sec. F singular
“Find $\text{adj } A$ ”	Sec. F cofactor/transpose
“Find A^{-1} ” directly	Sec. G, $A^{-1} = \frac{1}{ A } \text{adj } A$
Given $A^2 =$ linear combo of A, I ; find A^{-1}	Worked Pattern: Inverse from Polynomial
“Solve using matrices” with 3 linear equations	Worked Pattern: $AX = B$
“Check if consistent”	Sec. I — test $ A $, then $(\text{adj } A)B$
System $= 0$ on the right (homogeneous)	Sec. J trivial/non-trivial
$(AB)^{-1}$, $\text{adj}(AB)$ in a proof	Sec. H reversal laws

CHAPTER 5

Continuity & Differentiability

Formula Reference Sheet

ISC Class XII Maths — Chapter 5

Continuity & Differentiability — Formula Sheet

A. Continuity

f is continuous at $x = c \iff \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$, i.e. LHL = RHL = $f(c)$.

$$\text{LHL} = \lim_{h \rightarrow 0} f(c - h) \quad \text{RHL} = \lim_{h \rightarrow 0} f(c + h)$$

Algebra of continuous functions: if f, g are continuous at c , so are $f \pm g$, fg , kf , and f/g (where $g(c) \neq 0$); the composite $f(g(x))$ is continuous wherever g is continuous at c and f is continuous at $g(c)$.

$$|x - a| = \begin{cases} x - a, & x \geq a \\ -(x - a), & x < a \end{cases} \text{ — split before taking one-sided limits.}$$

B. Differentiability

$$Lf'(c) = \lim_{h \rightarrow 0^-} \frac{f(c + h) - f(c)}{h}$$

$$Rf'(c) = \lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h}$$

f is differentiable at $c \iff Lf'(c) = Rf'(c)$ (both finite and equal).

Differentiable $\Rightarrow$ continuous, but **not conversely** — e.g. $f(x) = |x|$ is continuous but not differentiable at 0.

C. Standard Trigonometric Limits

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{\tan \theta}{\theta} = 1$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta} = 0$$

$$\lim_{\theta \rightarrow 0} \frac{1 - \cos \theta}{\theta^2} = \frac{1}{2}$$

Tip: Always manipulate the given limit algebraically until it matches $\frac{\sin(\text{something} \rightarrow 0)}{\text{that same something}}$ before applying the standard result — multiply/divide to create matching angles, as in $\lim_{x \rightarrow 0} \frac{x}{\sin 3x} = \lim_{x \rightarrow 0} \frac{1}{3} \cdot \frac{3x}{\sin 3x}$.

D1. Pythagorean & Sum/Diff

Pythagorean:

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Sum/Difference:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

D2. Double & Half Angle

Double angle:

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Half-angle (for simplification):

$$1 - \cos x = 2 \sin^2 \frac{x}{2} \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

$$1 \pm \sin x = \left(\cos \frac{x}{2} \pm \sin \frac{x}{2} \right)^2$$

D3. Sum-to-Product

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

D4. Co-function Identities

Using $\left(\frac{\pi}{2} - x\right)$:

$$\sin x = \cos\left(\frac{\pi}{2} - x\right)$$

$$\tan x = \cot\left(\frac{\pi}{2} - x\right)$$

$$\sec x = \csc\left(\frac{\pi}{2} - x\right)$$

E1. Derivatives — Algebraic, Exp, Log

$$\frac{d}{dx}(x^n) = nx^{n-1} \quad \frac{d}{dx}(k) = 0$$

$$\frac{d}{dx}(e^x) = e^x \quad \frac{d}{dx}(a^x) = a^x \ln a$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

E2. Derivatives — Trigonometric

$$\frac{d}{dx} \sin x = \cos x \quad \frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x \quad \frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x \quad \frac{d}{dx} \csc x = -\csc x \cot x$$

E3. Derivatives — Inverse Trig

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} \quad \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}} \quad \frac{d}{dx} \csc^{-1} x = \frac{-1}{|x|\sqrt{x^2-1}}$$

F. Chain, Product & Quotient Rules

Chain rule: if $y = f(u)$, $u = g(x)$: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Product rule: $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

Quotient rule: $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

Inverse-Trig via Substitution

Before differentiating $\sin^{-1}/\cos^{-1}/\tan^{-1}$ of a messy expression, substitute to collapse it to a single angle:

$$x = \sin \theta \Rightarrow \sqrt{1-x^2} = \cos \theta$$

$$x = \tan \theta \Rightarrow \sqrt{1+x^2} = \sec \theta$$

$$x = \cos \theta \Rightarrow \sqrt{1-x^2} = \sin \theta$$

For $\sqrt{\frac{1-\cos x}{1+\cos x}}$ -type: use half-angle (Sec. D2) to reduce to $\tan \frac{x}{2}$, so $\tan^{-1}(\cdot) = \frac{x}{2}$, giving $\frac{dy}{dx} = \frac{1}{2}$ instantly.

G. Implicit Differentiation

Differentiate *both sides* w.r.t. x , treating y as a function of x : every term in y picks up a factor $\frac{dy}{dx}$ (chain rule), e.g. $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$.

Then collect all $\frac{dy}{dx}$ terms on one side and solve.

Logarithmic Differentiation

Use when the function is a **product/quotient of many factors**, or has a **variable in both base and exponent** (u^v):

1. Take $\ln$ of both sides: $\ln y = \ln(\text{RHS})$.
2. Expand using $\ln(ab) = \ln a + \ln b$, $\ln(a/b) = \ln a - \ln b$, $\ln(a^n) = n \ln a$.
3. Differentiate implicitly: LHS gives $\frac{1}{y} \frac{dy}{dx}$.
4. Multiply through by y (the original function) to isolate $\frac{dy}{dx}$.

H. Parametric Differentiation

$$\text{If } x = f(t), y = g(t): \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Second derivative: $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$ — differentiate $\frac{dy}{dx}$ w.r.t. t again, then divide by dx/dt (never divide by d^2x/dt^2).

I. Function w.r.t. Another Function

To find $\frac{dy}{du}$ where both y, u are functions of x : $\frac{dy}{du} = \frac{dy/dx}{du/dx}$

Reciprocal rule: $\frac{dy}{dx} = \frac{1}{dx/dy}$ — useful when x is easier to isolate than y (isolate x in terms of y , differentiate, then reciprocate).

Infinite Recursive Forms

For $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + \cdots}}}$ (∞):

1. Use self-similarity: the expression under the first root is the whole thing again, so $y = \sqrt{f(x) + y}$.
2. Square both sides: $y^2 = f(x) + y$.
3. Differentiate implicitly and solve for $\frac{dy}{dx}$.

Same idea for infinite continued fractions/powers — rewrite using self-similarity first, always *before* differentiating.

J. Second-Order Derivatives

$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right)$ — differentiate $\frac{dy}{dx}$ again w.r.t. x (product/chain rule as needed).

For implicit relations: find $\frac{dy}{dx}$ first (implicitly), then differentiate that expression again implicitly — substitute the first-derivative result back in to simplify $\frac{d^2y}{dx^2}$ fully in terms of x, y only.

K. Common Traps to Avoid

- Continuity needs **all three** equal: LHL, RHL, *and* $f(c)$ — matching limits alone isn't enough if $f(c)$ differs.
- Differentiability $\Rightarrow$ continuity, but the converse is false — $|x|$ is the classic counterexample.
- In logarithmic differentiation, don't forget to multiply back by the *original* y at the end.
- Parametric $\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{d^2x/dt^2}$ — must differentiate dy/dx w.r.t. t , then divide by dx/dt .
- Every term containing y needs a $\frac{dy}{dx}$ factor in implicit differentiation — easy to drop on product terms like xy (needs product rule: $x \frac{dy}{dx} + y$).

L. Quick Recognition Guide

Pattern seen	Use
Piecewise function, “examine continuity at $x = c$ ”	Sec. A — LHL, RHL, $f(c)$
$\frac{0}{0}$ form with $\sin/\tan$ as $x \rightarrow 0$, find k	Sec. C standard trig limits
$ x-a $ inside the function	Sec. A — split into cases first
“Is f derivable/differentiable at...”	Sec. B, $Lf' = Rf'$
$\tan^{-1}/\cos^{-1}$ of a messy fraction before “differentiate”	Worked Pattern: Inverse-Trig Substitution
$\sqrt{\frac{1 - \cos x}{1 + \cos x}}$ or similar inside inverse trig	Sec. D2 half-angle first
Product of 3+ factors, or u^v form	Worked Pattern: Logarithmic Differentiation
$x = f(t)$, $y = g(t)$ given	Sec. H parametric
Equation mixing x, y with no easy $y = \dots$ form	Sec. G implicit
“Find $\frac{dy}{du}$ ” with both in terms of x	Sec. I
Infinite root/fraction “ $\dots\infty$ ”	Worked Pattern: Infinite Recursive Forms
Asked for $\frac{d^2y}{dx^2}$	Sec. J (direct or implicit)

CHAPTER 6

Application of Derivatives

Formula Reference Sheet

ISC Class XII Maths — Chapter 6

Application of Derivatives — Formula Sheet

A. Rate of Change — Core Principle

If $y = f(x)$ and both x, y vary with time t , then by the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

More generally, if a relation connects several time-varying quantities (e.g. V, r, h), differentiate the relation *implicitly with respect to t* to connect their rates.

A rate is **positive** if the quantity is increasing, **negative** if decreasing — always state the sign in the final answer.

B1. Mensuration — Circle, Square, Cube

Circle (radius r): Area $A = \pi r^2$ Circumference $C = 2\pi r$

Square (side a): Area $A = a^2$ Perimeter $P = 4a$

Rectangle (sides l, b): Area $A = lb$ Perimeter $P = 2(l + b)$

Cube (side a): Volume $V = a^3$ Surface area $S = 6a^2$

B2. Mensuration — Sphere, Cylinder, Cone

Sphere (radius r): Volume $V = \frac{4}{3}\pi r^3$ Surface area $S = 4\pi r^2$

Cylinder (radius r , height h): Volume $V = \pi r^2 h$
Curved S.A. = $2\pi r h$ Total S.A. = $2\pi r h + 2\pi r^2$

Cone (radius r , height h , slant l): Volume $V = \frac{1}{3}\pi r^2 h$
 $l^2 = r^2 + h^2$ Curved S.A. = $\pi r l$

Cone with semi-vertical angle α : $\tan \alpha = \frac{r}{h}$, so $r = h \tan \alpha$ — substitute this to write V in terms of h alone.

Related Rates Problems

1. Identify all time-varying quantities and name them ($r, h, V, \dots$); note the given rate(s) and the rate asked for.
2. Write the geometric relation connecting the quantities (Sec. B1/B2), reducing to **one** independent variable using any given constraint (e.g. constant ratio, semi-vertical angle).
3. Differentiate the relation *implicitly with respect to t* .
4. Substitute the known values (including the value of the variable itself, found if necessary) and solve for the required rate.

Similar Triangles Setup

For a light source and a walking person, or height vs. horizontal distance problems (shadow-length, kite-string type):

1. Draw the figure; identify two similar right triangles (or one right triangle via Pythagoras for kite-string problems).
2. Write the proportion between corresponding sides (similar triangles), or $x^2 + h^2 = s^2$ (Pythagoras, kite at constant height h).
3. Differentiate with respect to t ; substitute known rates and values to solve.

C. Tangents & Normals — Core Formulas

Slope of tangent at (x_1, y_1) : $m_T = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}$

Slope of normal: $m_N = -\frac{1}{m_T}$ (normal $\perp$ tangent)

Equation of tangent: $y - y_1 = m_T(x - x_1)$

Equation of normal: $y - y_1 = m_N(x - x_1)$

Angle of inclination θ with the x -axis: $m = \tan \theta$

D. Tangents & Normals — Special Conditions

Tangent parallel to x -axis: $\frac{dy}{dx} = 0$ **Tangent parallel to y -axis** (normal horizontal): $\frac{dx}{dy} = 0$

Parallel to a line of slope m : $m_T = m$ **Perpendicular to a line** of slope m : $m_T \cdot m = -1$

Angle between two curves at a common point:
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Orthogonal intersection: $m_1 \cdot m_2 = -1$

Two curves touch at a point $\iff$ they have the same y -value **and** the same slope there.

Tangent & Normal Equations

1. If only x_1 (or only y_1) is given, substitute into the curve's equation to find the other coordinate.
2. Differentiate the curve to get $\frac{dy}{dx}$; evaluate at (x_1, y_1) to get m_T (implicit differentiation if the curve mixes x, y ; parametric formula $\frac{dy/dt}{dx/dt}$ if parametric).
3. Find $m_N = -1/m_T$.
4. Write both equations using $y - y_1 = m(x - x_1)$ with the respective slope.

E. Increasing & Decreasing Functions

f is **increasing** on interval I if $f'(x) \geq 0$ for all $x \in I$ (**strictly increasing** if $f'(x) > 0$).

f is **decreasing** on I if $f'(x) \leq 0$ for all $x \in I$ (**strictly decreasing** if $f'(x) < 0$).

Method: find $f'(x)$, factorise fully, then solve $f'(x) \geq 0$ and $f'(x) \leq 0$ using the sign scheme.

Wavy Curve Method

To solve $f'(x) \geq 0$ where $f'(x)$ is a product/quotient of linear (or repeated) factors:

1. Factorise $f'(x)$ completely; mark all zeros of numerator *and* denominator on a number line (denominator zeros are always excluded from the domain).
2. Starting from the right of the largest root, mark +; cross each root right-to-left, **flipping** sign at roots of odd multiplicity, and **not flipping** at roots of even multiplicity (repeated/squared factors).
3. Read off the intervals where $f'(x) > 0$ (increasing) and $f'(x) < 0$ (decreasing) directly from the sign pattern.

F. Maxima & Minima — Critical Points & Tests

Critical (stationary) point: $f'(c) = 0$ or $f'(c)$ does not exist.

First Derivative Test: if f' changes sign $+$ $\rightarrow$ $-$ at c , local **max**; if $- \rightarrow +$, local **min**; if no sign change, neither.

Second Derivative Test: at a point where $f'(c) = 0$: if $f''(c) < 0 \Rightarrow$ local **max**; if $f''(c) > 0 \Rightarrow$ local **min**; if $f''(c) = 0 \Rightarrow$ test fails, fall back to the first derivative test.

G. Absolute Maxima & Minima on $[a, b]$

1. Find all critical points of f inside the open interval (a, b) (solve $f'(x) = 0$).
2. Evaluate f at *every* critical point found **and** at both endpoints $f(a), f(b)$.
3. The **largest** value found is the absolute maximum; the **smallest** is the absolute minimum.

Tip: No second derivative test is needed for absolute extrema on a closed interval — just compare the raw function values from step 2 directly.

H. Common Traps to Avoid

- Every related-rates problem needs **implicit differentiation w.r.t. t** — never differentiate w.r.t. the spatial variable by mistake.
- Reduce to *one* variable using the given constraint *before* differentiating in optimization — differentiating a two-variable expression directly is a common error.
- Always check the **domain**: reject negative lengths/radii, and confirm a critical point actually lies inside the valid physical range.
- For "two curves touch," you need *both* equal y -values *and* equal slopes at the point — equal y alone only gives intersection, not tangency.
- In the wavy-curve method, a **repeated (even-power) factor** does not change sign at its root — forgetting this flips the entire sign pattern from that point onward.
- $f''(c) = 0$ does **not** mean "no extremum" — it means the second derivative test is inconclusive; go back to the first derivative test.

General Optimization Strategy

1. Read carefully and assign variable names to every quantity; note what is to be maximized/minimized and what is held **constant** (the constraint).
2. Use the constraint to eliminate one variable, writing the quantity to optimize as a function of a **single** variable (use Sec. B1/B2 for standard shapes).
3. Differentiate, set $= 0$, and solve for the critical value(s) within the valid physical domain (reject negative lengths, etc.).
4. Apply the second derivative test (Sec. F) to confirm max or min — or compare boundary values if the domain is a closed interval.
5. Substitute back to state the actual maximum/minimum value **and**, if asked, the dimensions that achieve it.

I. Quick Recognition Guide

Pattern seen	Use
“Rate of change of...”, two quantities linked by a formula, one rate given	Sec. A + Worked Pattern: Related Rates
Standard shape (circle, cube, sphere, cylinder, cone) dimensions changing with time	Sec. B1/B2 for the relevant formula, then differentiate w.r.t. t
Ladder, shadow, kite, or walking-person problems	Worked Pattern: Similar Triangles
“Find the slope of the tangent/normal at...”	Sec. C
“Find the equation of the tangent/normal...”	Worked Pattern: Tangent & Normal Equations
Tangent parallel/perpendicular to a given line or axis	Sec. D
Two curves given, “find angle between them” or “show orthogonal”	Sec. D angle/orthogonal formula
“Show the curves touch”	Sec. D — match y and slope
“Show f is increasing/decreasing” or “find intervals of monotonicity”	Sec. E + Worked Pattern: Wavy Curve
“Find local maxima/minima”	Sec. F, first or second derivative test
“Find absolute max/min on $[a, b]$ ”	Sec. G — compare all critical points <i>and</i> endpoints
Word problem: maximize/minimize area, volume, cost, perimeter subject to a constraint	Worked Pattern: General Optimization Strategy

CHAPTER 7

Integrals

Formula Reference Sheet

ISC Class XII Maths — Chapter 7

Integrals — Formula Sheet

A. Basic Standard Integrals

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \quad \int \frac{1}{x} dx = \log|x| + C$$

$$\int e^x dx = e^x + C \quad \int a^x dx = \frac{a^x}{\log a} + C$$

Linear-argument versions: $\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)} + C$

$$\int \frac{dx}{ax + b} = \frac{1}{a} \log|ax + b| + C$$

$$\int e^{kx} dx = \frac{1}{k} e^{kx} + C$$

B. Standard Trig Integrals

$$\int \sin x dx = -\cos x + C \quad \int \cos x dx = \sin x + C$$

$$\int \sec^2 x dx = \tan x + C \quad \int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C \quad \int \csc x \cot x dx = -\csc x + C$$

$$\int \tan x dx = \log|\sec x| + C \quad \int \cot x dx = \log|\sin x| + C$$

$$\int \sec x dx = \log|\sec x + \tan x| + C \quad \int \csc x dx = \log|\csc x - \cot x| + C$$

C. Six Special Rational/Root Forms

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log\left|\frac{x-a}{x+a}\right| + C$$

$$\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log\left|\frac{a+x}{a-x}\right| + C$$

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{dx}{\sqrt{x^2 - a^2}} = \log\left|x + \sqrt{x^2 - a^2}\right| + C$$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \log\left|x + \sqrt{x^2 + a^2}\right| + C$$

Tip: These six are the destination of almost every rational/irrational integral in this chapter — the whole game is manipulating the given integral (completing the square, splitting the numerator) until it matches one of these six exactly.

D. Three $\sqrt{\cdot}$ Integral Forms

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log\left|x + \sqrt{x^2 - a^2}\right| + C$$

$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log\left|x + \sqrt{x^2 + a^2}\right| + C$$

Integration by Substitution

Trigger: the integrand contains a function *and* (a constant multiple of) its own derivative.

1. Spot the inner function $g(x)$ whose derivative $g'(x)$ (or a constant multiple) also appears.

2. Let $u = g(x) \Rightarrow du = g'(x) dx$.

3. Rewrite the entire integral in terms of u only, integrate using Sec. A/B, then back-substitute $u = g(x)$.

Common substitution triggers: $x^n \rightarrow u = x^n$ (when $x^{n-1}dx$ present); $e^x \rightarrow u = e^x$; $\log x \rightarrow u = \log x$ (when $\frac{1}{x}dx$ present); $\sin x \rightarrow u = \sin x$ (when $\cos x dx$ present, and vice versa); $\tan x \rightarrow u = \tan x$ (when $\sec^2 x dx$ present).

E. Integration by Parts

$$\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$$

ILATE rule (priority order for choosing u): **I**nverse trig $\rightarrow$ **L**ogarithmic $\rightarrow$ **A**lgebraic $\rightarrow$ **T**rigonometric $\rightarrow$ **E**xponential — whichever appears *first* in this order is taken as u ; the rest is v .

F. Special By-Parts Results

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

$$\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + C$$

$$\int e^{ax} \cos(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \cos bx + b \sin bx) + C$$

Tip: The $e^x[f(x) + f'(x)]$ form is instantly recognisable: two terms added inside, one of which is visibly the derivative of the other — the answer is always just $e^x \times$ (the original $f(x)$), no actual by-parts work needed once spotted.

Partial Fractions

Use whenever integrating $\frac{P(x)}{Q(x)}$ with $Q(x)$ factorisable and $\deg P < \deg Q$ (if $\deg P \geq \deg Q$, divide first — Sec. below).

Distinct linear factors: $\frac{px+q}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$

Repeated linear factor: $\frac{1}{(x-a)^2(x-b)} = \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$

Irreducible quadratic factor: $\frac{1}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$

If $\deg P \geq \deg Q$: divide first, $\frac{P(x)}{Q(x)} = \text{Quotient} + \frac{\text{Remainder}}{Q(x)}$, then apply partial fractions to the remainder term only.

Linear-Numerator Integrals

For $\int \frac{px+q}{ax^2+bx+c} dx$ or $\int \frac{px+q}{\sqrt{ax^2+bx+c}} dx$ (numerator is linear, denominator quadratic):

1. Write $px+q = A \cdot \frac{d}{dx}(ax^2+bx+c) + B$, i.e. $px+q = A(2ax+b) + B$.
2. Compare coefficients to solve for A, B .
3. Split into two integrals: the $A(2ax+b)$ part integrates directly (it's exactly the derivative of the denominator, giving a log or power); the constant B part reduces to one of the six standard forms (Sec. C) after completing the square.

G. Trig Integrand Techniques

$\sin^2 x = \frac{1 - \cos 2x}{2}$ $\cos^2 x = \frac{1 + \cos 2x}{2}$ — for even powers, rewrite before integrating.

$\sin^3 x = \sin x(1 - \cos^2 x)$, then let $t = \cos x$ — for odd powers, peel off one factor and substitute.

Products like $\sin x \sin 2x \sin 3x$: convert to a sum first using Product-to-Sum identities, *then* integrate term by term.

Full identity toolkit (double/triple angle, product-to-sum, Weierstrass): see the **Master Trigonometry Reference**.

H. Definite Integral Properties

$$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ if } f \text{ is even } (f(-x) = f(x)); = 0 \text{ if } f \text{ is odd } (f(-x) = -f(x))$$

$$\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx \text{ if } f(2a-x) = f(x); = 0 \text{ if } f(2a-x) = -f(x)$$

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx \text{ (additivity, any } c)$$

I. Common Traps to Avoid

- Never forget the constant $+C$ for indefinite integrals — it's routinely worth a mark and just as routinely dropped when rushing.
- In Sec. C, the three log-form results look almost identical — double-check which of $x^2 - a^2$, $a^2 - x^2$ is in *your* denominator before quoting the formula; they are not interchangeable.
- For $\int \frac{px+q}{ax^2+bx+c} dx$, always split into the derivative-of-denominator part *and* the constant part — integrating the whole numerator directly against the denominator is a common but invalid shortcut.
- When using Weierstrass substitution, don't forget to convert dx itself (Master Trig Reference) — substituting only $\sin x, \cos x$ and leaving dx unconverted is the most common slip on the hardest Ch. 7 integrals.
- For definite integrals using King's Property ($\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$), always add the original and transformed integrals together at the end — forgetting this final addition step leaves the answer as $2I$ instead of I .

J. Quick Recognition Guide

Pattern seen	Use
A function and its derivative both appear in the integrand	Worked Pattern: Substitution
Denominator is $x^2 \pm a^2$ or $a^2 - x^2$ (or $\sqrt{\cdot}$ of these)	Sec. C (memorise all six)
$\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$, or $\sqrt{x^2 + a^2}$ <i>alone</i> being integrated (no other x factor)	Sec. D
Product of two unlike functions (e.g. $x \log x$, $x \sin x$, $\tan^{-1} x$)	Sec. E, use ILATE to choose u
e^x multiplying a sum of a function and its derivative	Sec. F special result — instant answer
Rational function, denominator factorises into linear/quadratic pieces	Worked Pattern: Partial Fractions
Numerator is linear, denominator is a quadratic (or its square root)	Worked Pattern: Linear-Numerator
Even power of $\sin x / \cos x$ (like $\sin^2 x$, $\cos^4 x$)	Sec. G power-reduction first
Odd power of $\sin x / \cos x$	Sec. G peel off one factor, substitute
Denominator is purely $a + b \sin x$ or $a + b \cos x$	Weierstrass substitution (Master Trig Reference)
Definite integral with symmetric limits ($-a$ to a , or 0 to $2a$)	Sec. H even/odd properties
Definite integral where $f(a + b - x)$ looks related to $f(x)$	Sec. H King's Property

CHAPTER 8

Application of Integrals

Formula Reference Sheet

ISC Class XII Maths — Chapter 8

Application of Integrals — Formula Sheet

A. Area Under a Curve — Core Formula

Vertical strips (curve given as $y = f(x)$), area between the curve and the x -axis from $x = a$ to $x = b$:

$$\text{Area} = \int_a^b y \, dx = \int_a^b f(x) \, dx$$

Horizontal strips (curve given as $x = g(y)$), area between the curve and the y -axis from $y = c$ to $y = d$:

$$\text{Area} = \int_c^d x \, dy$$

Choose whichever strip makes the boundary equation simplest to integrate — vertical strips for "bounded by $x = a, x = b$," horizontal strips for "bounded by $y = c, y = d$."

B. Area Between Two Curves

For two curves with $y_{\text{upper}} \geq y_{\text{lower}}$ on $[a, b]$:

$$\text{Area} = \int_a^b (y_{\text{upper}} - y_{\text{lower}}) \, dx$$

If the curves cross inside $[a, b]$, split the integral at each crossing point and always subtract the lower curve from the upper curve on *that* sub-interval.

C. The Signature Standard Integral

$$\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$$

This single result drives almost every circle and ellipse area problem in this chapter — memorise it exactly, including the $\frac{1}{2}$ and $\frac{a^2}{2}$ coefficients.

D. Standard Curve Equations Reference

Circle (centre origin, radius a): $x^2 + y^2 = a^2 \Rightarrow y = \sqrt{a^2 - x^2}$

Parabola (opens right): $y^2 = 4ax \Rightarrow y = \sqrt{4ax}$
(opens up): $x^2 = 4ay$

Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow y = \frac{b}{a} \sqrt{a^2 - x^2}$

Line through $(x_1, y_1), (x_2, y_2)$: $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Area of a Circle

For $x^2 + y^2 = a^2$, use symmetry across all four quadrants rather than integrating the full circle directly:

1. Solve for $y = \sqrt{a^2 - x^2}$ (upper half, first quadrant arc).
2. Area of one quadrant = $\int_0^a \sqrt{a^2 - x^2} \, dx$; apply the standard result (Sec. C).
3. Multiply by 4 for the full circle: $\text{Area} = 4 \times \frac{\pi a^2}{4} = \pi a^2$.

Area of an Ellipse

For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

1. Solve for $y = \frac{b}{a} \sqrt{a^2 - x^2}$ (first-quadrant arc).
2. Area of one quadrant = $\int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx = \frac{b}{a} \cdot \frac{\pi a^2}{4}$ (standard result, Sec. C).
3. Multiply by 4: $\text{Area} = 4 \times \frac{\pi ab}{4} = \pi ab$ — quote this directly once derived once.

E. Circular Segments

A **segment** is the region cut off by a chord (a line not through the centre).

Minor segment (smaller piece): Area = (Area under the arc, from the chord's x to a) computed directly via Sec. C over the appropriate limits — draw the figure to identify the correct limits of integration.

Alternative: Segment area = Sector area – Triangle area, where Sector = $\frac{1}{2}a^2\theta$ and Triangle = $\frac{1}{2}a^2\sin\theta$ (θ in radians, the angle subtended at the centre).

Line-and-Curve Regions

1. Sketch both the line and the curve — this chapter's marks depend heavily on a correct rough figure.
2. Find point(s) of intersection by solving the two equations simultaneously.
3. Identify which curve is "on top" (y_{upper}) in each relevant sub-interval.
4. Set up Area = $\int (y_{\text{upper}} - y_{\text{lower}}) dx$ over each sub-interval (split at intersection points if the upper/lower curve swaps), and add the pieces.

F. Symmetry Shortcuts

If the region is symmetric about the x -axis: find the area of the **upper half only** and multiply by 2.

If symmetric about **both** axes (full circle/ellipse centred at origin): find the **first-quadrant** area and multiply by 4.

Always state *why* the symmetry holds (e.g. "the curve contains only even powers of y , so it is symmetric about the x -axis") before applying the multiplier — this is a method mark in ISC board answers.

G. Common Traps to Avoid

- Always **sketch the region first** — setting up limits without a figure is the single biggest source of wrong answers in this chapter.
- In Sec. C's standard result, don't drop the $\frac{1}{2}$ and $\frac{a^2}{2}$ coefficients, and don't forget the $\sin^{-1}$ term entirely (a very common slip when rushing).
- When two curves swap which one is "on top," you **must** split the integral at the crossing point — a single integral across the whole interval silently gives the wrong (signed, cancelling) answer.
- Area is always reported as a **positive** quantity — if a computed integral comes out negative, take the absolute value; it means y_{lower} and y_{upper} were swapped.
- Double-check which variable you integrated with respect to matches the strip orientation you sketched (vertical strips $\rightarrow dx$, horizontal strips $\rightarrow dy$).

H. Quick Recognition Guide

Pattern seen	Use
“Area bounded by $y = f(x)$ and lines $x = a, x = b$ ”	Sec. A vertical strips
“Area bounded by ... and $y = c, y = d$ ”	Sec. A horizontal strips
Two curves/lines given, “find the area between/enclosed by”	Sec. B, find intersection first
Circle equation $x^2 + y^2 = a^2$ appears anywhere in the integral	Sec. C standard result
“Find the area of the circle/ellipse” (whole shape)	Worked Pattern: Circle or Ellipse by Symmetry
“Find the area of the (minor/major) segment”	Sec. E
Line <i>and</i> curve both given in the same question	Worked Pattern: Line and Curve
Region symmetric about an axis (even powers only)	Sec. F — halve the work, double/quadruple the result
Answer comes out negative	Sec. G — take absolute value, check which curve is on top

BONUS SECTION
Master Trigonometry Reference

Every Trig Identity Used in Chapters 1–8

Plus standard reference formulas and exam tricks

ISC Class XII Mathematics

Master Trigonometry Reference — All Chapters (1–8)

Every trig identity used across the syllabus, plus standard reference formulas and exam tricks

A. Basic Ratios & Reciprocal Relations

$$\sin \theta = \frac{1}{\csc \theta} \quad \cos \theta = \frac{1}{\sec \theta} \quad \tan \theta = \frac{1}{\cot \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Used everywhere — converting between ratios is the first step in most simplification problems (Ch. 2, 5, 7).

B. Pythagorean Identities

$$\sin^2 \theta + \cos^2 \theta = 1 \quad 1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \csc^2 \theta$$

Rearranged forms (used constantly in integration, Ch. 7):
 $1 - \sin^2 \theta = \cos^2 \theta \quad 1 - \cos^2 \theta = \sin^2 \theta$
 $\tan^2 \theta = \sec^2 \theta - 1 \quad \cot^2 \theta = \csc^2 \theta - 1$

C. Standard Angle Values

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	—	0

★ Quadrant Sign Rule (ASTC)

Going anticlockwise from Quadrant I: **All** positive, **Sin** (& csc) positive, **Tan** (& cot) positive, **Cos** (& sec) positive — everything else in that quadrant is negative.

Mnemonic: “Add Sugar To Coffee” or “All Students Take Chemistry” — Q1 All, Q2 Sin, Q3 Tan, Q4 Cos.

D. Co-function Identities

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta \quad \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta \quad \cot\left(\frac{\pi}{2} - \theta\right) = \tan \theta$$

$$\sec\left(\frac{\pi}{2} - \theta\right) = \csc \theta \quad \csc\left(\frac{\pi}{2} - \theta\right) = \sec \theta$$

E. Negative-Angle Identities

$$\sin(-\theta) = -\sin \theta \quad \cos(-\theta) = \cos \theta \quad \tan(-\theta) = -\tan \theta$$

Sine, tan, cot, cosec are **odd** (flip sign); cosine and sec are **even** (sign stays).

★ Quadrant Reduction Rule

For $(n \cdot 90^\circ \pm \theta)$: if n is **even** (i.e. $0^\circ, 180^\circ, 360^\circ$), the ratio's **name stays the same**. If n is **odd** (i.e. $90^\circ, 270^\circ$), the ratio **flips** ($\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \csc$). The **sign** is whatever that ratio would be in the resulting quadrant (use ASTC).

E.g. $\sin(180^\circ - \theta) = \sin \theta$ (name same, Q2 sine is +).
 $\cos(90^\circ + \theta) = -\sin \theta$ (name flips, Q2 cosine is -).

F. Sum & Difference Formulas

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

Used in: Ch. 3 (rotation matrices), Ch. 7 (integration by expansion).

G. Double Angle Formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$
 (all three forms are used — pick whichever matches what's already in the expression)

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad \cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$$

★ Teacher's Trick: Power Reduction

This is exactly the “ $1 - \cos 2\theta$ ” rule — used to rewrite $\sin^2 \theta$ or $\cos^2 \theta$ **before integrating**, since you can't directly integrate a squared trig ratio:

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2} \quad \cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

How to remember which sign goes where: sine is “small at 0,” so $\sin^2 0 = 0$ needs the *minus* form ($\frac{1-1}{2} = 0$); cosine is “big at 0,” so $\cos^2 0 = 1$ needs the *plus* form ($\frac{1+1}{2} = 1$).

Trigger: the moment you see $\sin^2 x$ or $\cos^2 x$ sitting inside an integral with no obvious substitution, reach for this first.

H. Half-Angle Formulas

$$1 - \cos x = 2 \sin^2 \frac{x}{2} \quad 1 + \cos x = 2 \cos^2 \frac{x}{2}$$

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$1 \pm \sin x = \left(\sin \frac{x}{2} \pm \cos \frac{x}{2} \right)^2$$

Used in: Ch. 5 & Ch. 7 for simplifying $\sqrt{\frac{1 - \cos x}{1 + \cos x}}$ type expressions before differentiating/integrating.

I. Triple Angle Formulas

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

Used in: Ch. 7, when integrands contain $\sin^3 x$ or $\cos^3 x$ — rearrange these to isolate $\sin^3 \theta$ or $\cos^3 \theta$.

J. Product-to-Sum Formulas

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

Tip: These turn a *product* of trig functions into a *sum* — essential whenever you need to integrate something like $\sin x \sin 2x \sin 3x$ or $\cos 4x \cos 2x$ (Ch. 7): a product can't be integrated directly, but each term of a sum can.

K. Sum-to-Product Formulas

$$\sin A + \sin B = 2 \sin \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A + B}{2} \sin \frac{A - B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A + B}{2} \cos \frac{A - B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A + B}{2} \sin \frac{A - B}{2}$$

Tip: The reverse direction of Sec. J — turns a *sum/difference* into a *product*, useful for simplifying numerators/denominators in Ch. 2 and Ch. 7 before cancelling common factors.

Weierstrass Substitution

The single most powerful substitution for integrals mixing $\sin x$ and $\cos x$ in a rational expression:

Let $t = \tan \frac{x}{2}$. Then:

$$\sin x = \frac{2t}{1 + t^2} \quad \cos x = \frac{1 - t^2}{1 + t^2} \quad dx = \frac{2 dt}{1 + t^2}$$

Trigger: an integral of the form $\int \frac{dx}{a + b \cos x}$ or $\int \frac{dx}{a + b \sin x}$ (linear combination of $\sin x, \cos x$ in the denominator, nothing else) — this is the standard recognition cue used throughout Ch. 7.

L. General Solutions of Trig Equations

$$\sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha, \quad n \in \mathbb{Z}$$

$$\cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha, \quad n \in \mathbb{Z}$$

$$\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha, \quad n \in \mathbb{Z}$$

Standard reference — not heavily used in Ch. 1–8 directly, but foundational background for inverse trig (Ch. 2).

M. Inverse Trig Quick Reference

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

$$\sec^{-1} x + \csc^{-1} x = \frac{\pi}{2}$$

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right), \quad xy < 1$$

Full treatment with all properties, right-triangle method, and worked patterns: see the **Chapter 2 formula sheet**.

N. Derivatives & Integrals Index

$$\frac{d}{dx} \sin x = \cos x \quad \frac{d}{dx} \cos x = -\sin x \quad \frac{d}{dx} \tan x = \sec^2 x$$

$$\int \sin x \, dx = -\cos x + C \quad \int \cos x \, dx = \sin x + C$$

$$\int \tan x \, dx = \log |\sec x| + C$$

Full tables (all 6 ratios, both directions): see the **Chapter 5** (derivatives) and **Chapter 7** (integrals) formula sheets.

★ Tips: Spotting the Right Formula

• **See $\sin^2 x$ or $\cos^2 x$ alone in an integral?** → Power-reduction (Sec. I): rewrite using $\frac{1 \mp \cos 2x}{2}$.

• **See $\sin^3 x$ or $\cos^3 x$?** → Split off one factor and use the Pythagorean identity (e.g. $\sin^3 x = \sin x(1 - \cos^2 x)$), *not* the triple-angle formula — that's for $\sin 3x$, a different thing entirely.

• **See $\sin x \cos x$ together as a product?** → Recognise it as $\frac{1}{2} \sin 2x$ instantly.

• **See a product of two different-angle trig terms** (like $\sin 4x \cos 2x$)? → Product-to-Sum (Sec. J).

• **Denominator is $a + b \cos x$ or $a + b \sin x$ only?** → Weierstrass substitution $t = \tan \frac{x}{2}$.

• **See $\sqrt{\frac{1 - \cos x}{1 + \cos x}}$ or similar under a root?** → Half-angle (Sec. H) collapses it to $\tan \frac{x}{2}$ (or $\cot, \csc$) directly — avoids messy algebra.

• **Angle is $90^\circ \pm \theta$ or $180^\circ \pm \theta$ etc.?** → Quadrant Reduction trick: name flips only for $90^\circ/270^\circ$, sign from ASTC.

• **Stuck simplifying a sum of two sines/cosines with different angles?** → Sum-to-Product (Sec. K) often reveals a common factor that cancels.

Common Mistakes to Avoid

• Mixing up **triple-angle** ($\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$) with **power-reduction** of $\sin^3 \theta$ — these solve different problems. Triple-angle expands $\sin(3\theta)$; power-reduction rewrites $(\sin \theta)^3$.

• In the quadrant reduction rule, forgetting that the ratio **name only flips at 90° and 270°** , not at 180° or 360° — a very common sign/name error.

• Writing $\cos 2\theta = 2 \cos \theta - 1$ instead of $2 \cos^2 \theta - 1$ — the square is easy to drop when writing quickly.

• Forgetting the $dx = \frac{2 \, dt}{1+t^2}$ conversion when using Weierstrass substitution — substituting $\sin x, \cos x$ correctly but leaving dx unconverted is the single most common slip in Ch. 7's hardest integrals.

• Applying co-function identities ($90^\circ - \theta$ forms) when the angle is actually $90^\circ + \theta$ — check the sign inside the bracket carefully; they are not interchangeable.

Where Each Formula Group Is Used

Formula group	Chapters that rely on it
Basic ratios & reciprocals (A)	Ch. 2, 5, 7 — everywhere
Pythagorean identities (B)	Ch. 2, 5, 6, 7
ASTC / quadrant signs (D–E)	Ch. 2 (principal values)
Sum/difference (F)	Ch. 3 (rotation matrices), Ch. 7
Double angle (G)	Ch. 2, 5, 6, 7 — the most-used identity group overall
Power reduction (teacher's trick)	Ch. 7 (integration of $\sin^2 x, \cos^2 x$)
Half-angle (H)	Ch. 5, 7 (simplification before diff./int.)
Triple angle (I)	Ch. 7 only ($\sin^3 x, \cos^3 x$ integrals)
Product-to-sum / sum-to-product (J–K)	Ch. 7 (products of sines/cosines)
Weierstrass substitution	Ch. 7 (rational trig integrals)
Inverse trig identities (M)	Ch. 2 entirely
Trig derivatives/integrals (N)	Ch. 5 (derivatives), Ch. 7 (integrals), Ch. 6 (applications)

Final Exam-Day Reminder

When a trig expression looks stuck, work through this checklist in order: (1) Can a Pythagorean identity simplify it? (2) Is there a double/half-angle pattern hiding? (3) Would Product-to-Sum or Sum-to-Product help? (4) For integration specifically — does Weierstrass substitution apply? Nine times out of ten, one of these four gets you moving again.